SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Scenario Based Questions

Course Code:	**DSA0404**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	CO4-AT1 -Scenario Based Questions	**Assessment:**	**Assessment Tool 1- Scenario Based Question**
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO4 Statement:	To apply the statistical inferences such as testing of hypothesis and point estimates for the given data		
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Code of Conduct
I Sania Herbert(192324062) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sania

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Que: A bank wants to estimate the average monthly expenditure of its credit- card customers. A random sample of 150 customers has a mean expenditure of ₹24,500. Explain how the sample mean can be used as a frequentist estimate of the population mean.

The information available is:

- Sample size:
- Sample mean expenditure:
- Population mean: — unknown

The objective is to explain how the sample mean can be used as a frequentist estimate of the population mean.

In this scenario:

- The population is all credit-card customers of the bank.
- The sample is the randomly selected 150 customers.
- The population mean is the true average monthly expenditure of all credit-card customers.
- The sample mean is the average expenditure calculated from the 150 selected customers.

Since the bank does not know the expenditure of every customer, the population mean cannot be directly calculated. Therefore, the bank uses the sample mean as an estimate. Thus, ₹24,500 is the point estimate of the population mean.

2. Population and Sample

Population

The population consists of all credit-card customers of the bank.

Let:μ=true average monthly expenditure of all customers

The μ value of is unknown.

Sample

A random sample of 150 customers is selected. N=150

The average expenditure of these customers is: $\bar{x} = ₹24,500$

Term	Meaning	Value
Population	All credit-card customers	All customers
Sample	Selected customers	150
Population parameter	True average expenditure	(\mu), unknown
Sample statistic	Average expenditure of sample	₹24,500
Estimator	Rule used to estimate (\mu)	(\bar{x})
Point estimate	Actual estimated value	₹24,500

3. What is the Frequentist Approach?

The frequentist approach is a method of statistical inference in which unknown population parameters are treated as fixed but unknown values, while sample statistics are treated as random quantities.

In this scenario:

μ=fixed but unknown

$\bar{x}$=random variable

because a different random sample of 150 customers could produce a different sample mean.

For example, suppose the bank repeatedly selects samples of 150 customers.

It might obtain:

Sample	Sample Mean
Sample 1	₹24,200
Sample 2	₹24,700
Sample 3	₹24,450
Sample 4	₹24,600
Sample 5	₹24,350

The sample mean changes from sample to sample.

However, the population mean μ remains fixed.

This repeated-sampling idea is the foundation of the frequentist approach.

4. Estimator and Estimate

There is an important difference between an **estimator** and an **estimate**.

Estimator

An estimator is a statistical rule or formula used to estimate an unknown parameter.

For estimating the population mean:

$\hat{\mu}=\bar{x}$

Thus, the **sample mean is the estimator** of the population mean.

Estimate

Once the sample data are observed, the estimator produces a particular numerical value.

Here: $\bar{x}$=₹24,500,μ=₹24,500

So:

- **Estimator:** $\bar{x}$
- **Estimate:** ₹24,500

5. Formula for Sample Mean

The sample mean is calculated as:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

where:

- = expenditure of customer
- = number of customers
- = sample mean

For this problem:

n=150, x¯=₹24,500

Therefore, the total expenditure represented by the sample would be:

$$\sum x_i = n\bar{x}$$

=150(24,500)
=₹36,75,000
So the 150 customers together have a total monthly expenditure of **₹36,75,000**, corresponding to an average of ₹24,500 per customer.

6. Point Estimation

A **point estimate** provides one numerical value as an estimate of an unknown population parameter.
The population parameter is:μ

The point estimator is: $\hat{\mu} = \bar{x}$

The observed point estimate is: $\boxed{\hat{\mu} = ₹24,500}$

Interpretation
The bank can report:
Based on the random sample of 150 credit-card customers, the estimated average monthly expenditure of the entire credit-card customer population is ₹24,500 per customer.
This does **not** mean every customer spends ₹24,500.
It means ₹24,500 is the estimated **population average**.

7. Why Can the Sample Mean Estimate the Population Mean?

The sample mean is widely used because it has desirable statistical properties.
The major properties are:

1. **Unbiasedness**
2. **Consistency**
3. **Efficiency**
4. **Minimum sampling variability under standard assumptions**

These properties make the sample mean a suitable frequentist estimator.

8. Property 1 – Unbiasedness

An estimator is unbiased if its expected value equals the true population parameter.

For the sample mean: $\boxed{E(\bar{x}) = \mu}$

This means that if we repeatedly select random samples of 150 customers and calculate their sample means, the average of those sample means will tend to equal the true population mean.

Example

Suppose repeated samples give: ₹24,200, ₹24,700, ₹24,400, ₹24,600, ₹24,600

Their average is: 24200+24700+24400+24600+24600/5

=122500/5

=₹24,500

This illustrates the idea of unbiasedness.

Therefore: $\boxed{E(\bar{x}) = \mu}$

Hence, the sample mean is an **unbiased estimator** of the population mean.

9. Property 2 – Consistency

An estimator is consistent if it approaches the true population parameter as the sample size increases.

For the sample mean: $\boxed{\bar{x} \to \mu \quad \text{as } n \to \infty}$

Therefore, as the bank increases its sample size, the sample mean generally becomes closer to the actual population mean.

For example:

- Sample size = 50 → less information
- Sample size = 150 → more information
- Sample size = 500 → even more information
- Sample size = 1,000 → greater precision

Thus, the sample mean is a **consistent estimator**.

10. Property 3 – Efficiency

The variance of the sample mean is:

$$\boxed{Var(\bar{x}) = \frac{\sigma^2}{n}}$$

where:

- = population variance
- = sample size

For this problem: Var(x¯)=σ2/150
As increases, the variance decreases.
Therefore, the sample mean becomes more stable across repeated samples.
This is an important reason why a sufficiently large sample is useful.

11. Standard Error of the Sample Mean

The standard error measures the sampling variability of the sample mean.
If the population standard deviation is known:

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}$$

Since: n=150
we have:

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{150}}$$

Since:

$$\sqrt{150} \approx 12.247$$

we get:

$$\boxed{SE(\bar{x}) \approx \frac{\sigma}{12.247}}$$

If the population standard deviation is unknown, which is common in practice, the sample standard deviation is used:

$$SE(\bar{x}) = \frac{s}{\sqrt{150}}$$

The question does not provide , so the exact standard error cannot be numerically calculated.

12. Central Limit Theorem

The sample size is:

which is sufficiently large for the **Central Limit Theorem (CLT)** to be useful under usual conditions.
The Central Limit Theorem states that the sampling distribution of the sample mean becomes approximately normal as the sample size becomes sufficiently large.
Therefore:

$$\boxed{\bar{x} \approx N\left(\mu, \frac{\sigma^2}{150}\right)}$$

This allows the bank to make statistical inferences about the population mean.
The CLT is particularly useful because the original expenditure distribution may not be perfectly normal.

13. Sampling Distribution

The **sampling distribution of the sample mean** represents the distribution of sample means obtained from repeated random samples of the same size.
Imagine the bank repeatedly selects 150 customers:
Sample 1→x¯1
Sample 2→x¯2
Sample 3→x¯3
⋮
Sample 1000→x¯1000
These sample means form a sampling distribution.
Its center is: E(x¯)=μ
and its variance is: Var(x¯)=σ2/n

14. Confidence Interval
A point estimate gives only one value.
A confidence interval provides a **range of plausible values** for the population mean.
If the population standard deviation is unknown, the 95% confidence interval can generally be written as:

$$\boxed{\bar{x} \pm t_{\alpha/2,n-1}\frac{s}{\sqrt{n}}}$$

For this problem:

$$\bar{x} = ₹24,500$$

$$n = 150$$

Therefore: CI95%=₹24,500±t0.025,149 s/150

Since is not provided, the exact numerical confidence interval cannot be calculated from the given information.

15. Numerical Illustration of Confidence Interval

To demonstrate the calculation, suppose the sample standard deviation was **₹6,000**.
This value is only an illustration because it was not given in the question.
Then: s=₹6,000,n=150
The standard error is: SE=6000/150
SE≈₹489.90
For approximately 149 degrees of freedom, the 95% t-critical value is approximately: t≈1.976
Therefore: CI=24,500±1.976(489.90)
Margin of error: 1.976×489.90≈₹968
Therefore: CI=₹24,500±₹968
Lower limit: ₹24,500−₹968=₹23,532
Upper limit: ₹24,500+₹968=₹25,468
Thus, the illustrative 95% confidence interval is:

$$\boxed{₹23,532 \leq \mu \leq ₹25,468}$$

Again, this interval is **illustrative only** because the actual sample standard deviation was not provided.

16. Correct Frequentist Interpretation of Confidence Interval

A common mistake is to say:
"There is a 95% probability that the population mean is between ₹23,532 and ₹25,468."
This is not the technically correct frequentist interpretation.
The correct interpretation is:
If the bank repeatedly selected random samples of 150 customers and constructed confidence intervals using the same procedure, approximately 95% of those intervals would contain the true population mean.
The population mean is treated as fixed.
The confidence interval is what varies from sample to sample.

17. Sampling Error

The sample mean may differ from the true population mean because the bank observes only a sample.

$$\boxed{\text{Sampling Error} = \bar{x} - \mu}$$

Sampling error can be represented as:
The actual value cannot be calculated because is unknown.
For example, if the true population mean were actually ₹24,700:

$$Sampling\ Error = 24,500 - 24,700$$

$$= -₹200$$

The sample would underestimate the population mean by ₹200.
This does not necessarily mean that the estimator is biased.

18. Random Sampling and Representativeness

The quality of the estimate depends heavily on how the sample is selected.
The bank should randomly select the 150 customers.
For example, it should avoid selecting only:

- High-income customers
- Premium credit-card customers
- Customers from one city
- Customers with very high credit limits

If the sample is not representative, the sample mean may be systematically different from the population mean.

Therefore: $$\text{Random sample} \rightarrow \text{Reliable inference}$$
while:

$$\text{Biased sample} \rightarrow \text{Potentially biased estimate}$$

19. Assumptions

For the frequentist estimation to be reliable, the following assumptions should be considered.

1. Random sampling
The 150 customers should be selected randomly.

2. Independence
The expenditure observations should be reasonably independent.

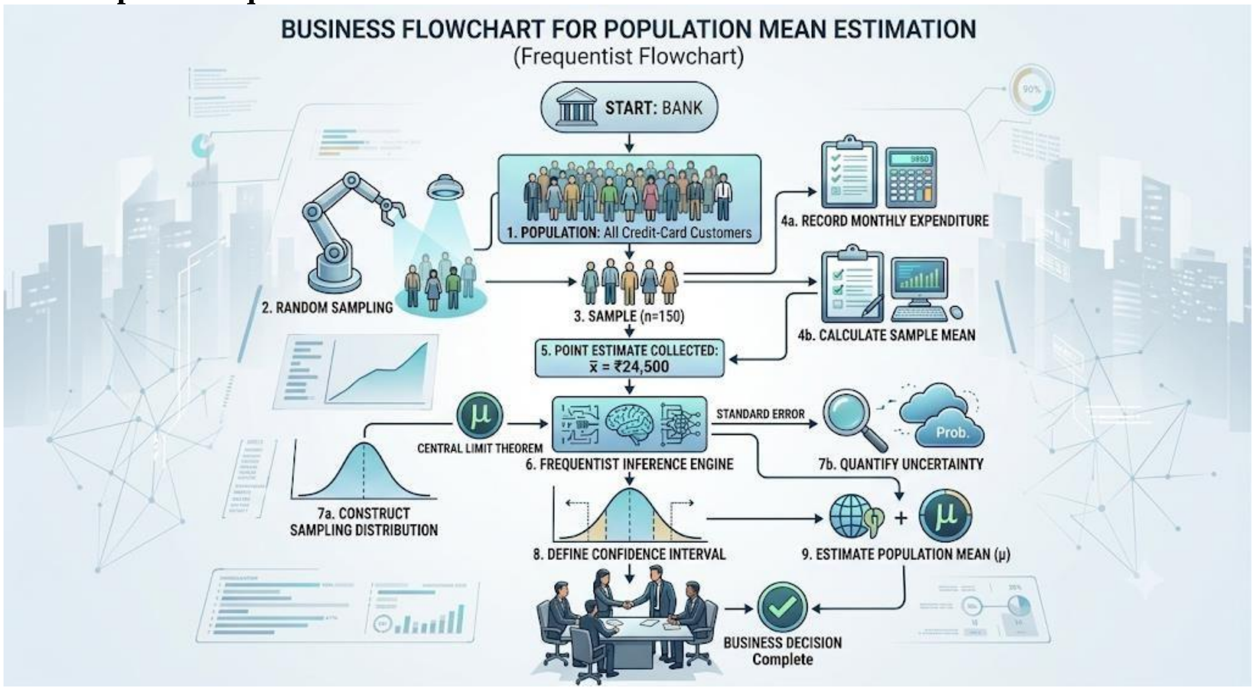
3. Representative sample
The sample should represent the target customer population.

4. Accurate measurements
The bank's expenditure data should be correctly recorded.

5. Finite variance
Customer expenditure should have a finite variance.

6. Adequate sample size
A sample size of 150 is sufficiently large for the Central Limit Theorem to generally support approximate normal inference.

23. Complete Frequentist Flowchart



26. Why Frequentist Rather Than Assuming the Exact Mean?

The bank does not know the true value of .
It therefore cannot simply state:μ=₹24,500
This distinction is important. ₹24,500 is a **sample-based estimate**, while is the unknown true population parameter.
The frequentist framework provides tools to determine how reliable this estimate is through:

- Sampling distributions
- Standard errors
- Confidence intervals
- Hypothesis tests
- Repeated-sampling properties